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Hydrogen Fuel Cells As A Renewable Source Of Energy

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I. Introduction

Since the industrial revolution, humans have depended on fossil fuels to produce energy. However, in recent decades, the impact of fossil fuels has become a major concern for global society. From air pollution to natural disasters, fossil fuels pose significant risks, necessitating the search for alternatives. Consequently, scientists worldwide are focusing on finding safe, environmentally friendly, and reliable energy supplies.

Over the years, various renewable energy solutions have emerged, such as solar panels, wind turbines, and hydropower. Among these, hydrogen energy, specifically hydrogen fuel cells, stands out as a particularly promising alternative to fossil fuels. Hydrogen fuel cells are clean, reliable, quiet, and efficient sources of high-quality electric power. They utilize hydrogen as fuel to drive an electrochemical process that produces electricity, with water and heat as the only by-products.

While hydrogen fuel cell technology is still in development and not yet implemented on a global scale, it offers significant potential for achieving zero emissions. However, challenges remain, particularly in the production of hydrogen itself. The "zero emission" premise relies on hydrogen being produced from renewable sources, such as splitting water molecules or using energy from solar panels and wind turbines. Currently, most hydrogen is still derived from natural gas and other non-renewable sources, which presents a significant hurdle to overcome. This presents an interesting study over its identity as a renewable energy, its efficiency, and the technology that drives it.

I.I Problem Identification

- Are hydrogen fuel cells a viable source of energy?
- How efficient are hydrogen fuel cells at generating power?
- Are hydrogen fuel cells truly a zero emission solution to fossil fuels?
- What are the drawbacks of hydrogen fuel cells?
- How do hydrogen fuel cells work in detail?

I.II Objective and Benefits

The objective of this study is to assess the viability of hydrogen fuel cells as a renewable energy source, by examining the advantages and disadvantages, as well as analyzing the system efficiency through the application of the energy conservation law. With this, the study could provide an evaluation of hydrogen fuel cells potential to contribute to sustainable energy solutions on a worldwide scale.

II. METHODOLOGY

Fuel Cell Underlying Concept

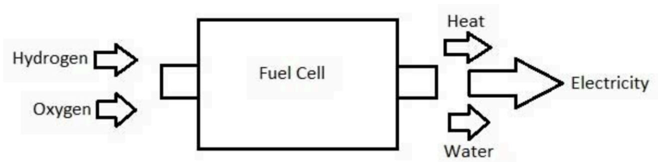

Figure 1. Fuel Cell Basic Mechanism Diagram

Fuel cells work like batteries, but they do not run down or need recharging. They produce electricity and heat as long as fuel is supplied. A fuel cell consists of two electrodes—a negative electrode (or anode) and a positive electrode (or cathode)—sandwiched around an electrolyte. A fuel, such as hydrogen gas is supplied to the anode of the fuel cell. The anode is coated with platinum, which acts as a catalyst to break down the hydrogen into protons and electrons. If a circuit is connected between the anode and cathode then the electrons can travel through the circuit and provide power to any load that is connected as part of the circuit.

Reaction at the anode (oxidation reaction):

$\mathbf{2H_2 \rightarrow 4H^+ + 4e^-}$

The flow of electrons through the load is the electric current that is generated by the fuel cell.

The hydrogen ions (protons) that are produced from the hydrogen at the anode travel through the electrolyte in the fuel cell to the cathode. Oxygen supplied to the cathode reacts with these hydrogen ions and electrons arriving via the external circuit to produce water and heat, both of which are removed from the fuel cell.

Reaction at the cathode (reduction reaction):

$\mathbf{O_2 + 4H^+ + 4e^- \rightarrow 2H_2O}$

Fuel Cell Types

Based on the type of electrolyte used, fuel cells can be categorized into alkaline fuel cells (AFCs), PEMFCs, phosphoric acid fuel cells (PAFCs), molten carbonate fuel cells (MCFCs), and solid oxide fuel cells (SOFCs). In this paper, we focus on hydrogen electrification through polymer electrolyte membrane fuel cells (PEMFCs), as it is a common type of fuel cell.

Parts of A Polymer Electrolyte Membrane Fuel Cells

PEMFCs are the current focus of research for fuel cell vehicle applications. PEM fuel cells are made from several layers of different materials. The heart of a PEM fuel cell is the membrane electrode assembly (MEA), which includes the membrane, the catalyst layers, and gas diffusion layers (GDLs). Hardware components used to incorporate an MEA into a fuel cell include gaskets, which provide a seal around the MEA to prevent leakage of gases, and bipolar plates, which are used to assemble individual PEM fuel cells into a fuel cell stack and provide channels for the gaseous fuel and air.

Membrane Electrode Assembly

The membrane, catalyst layers (anode and cathode), and diffusion media together form the membrane electrode assembly (MEA) of a PEM fuel cell.

Polymer Electrolyte Membrane

The polymer electrolyte membrane, or PEM (also called a proton exchange membrane)—a specially treated material that looks something like ordinary kitchen plastic wrap—conducts only positively charged ions and blocks the electrons. The PEM is the key to the fuel cell technology; it must permit only the necessary ions to pass between the anode and cathode. Other substances passing through the electrolyte would disrupt the chemical reaction. For transportation applications, the membrane is very thin—in some cases under 20 microns.

Catalyst Layers

A layer of catalyst is added on both sides of the membrane—the anode layer on one side and the cathode layer on the other. Conventional catalyst layers include nanometer-sized particles of platinum dispersed on a high-surface-area carbon support. This supported platinum catalyst is mixed with an ion-conducting polymer (ionomer) and sandwiched between the membrane and the GDLs. On the anode side, the platinum catalyst enables hydrogen molecules to be split into protons and electrons. On the cathode side, the platinum catalyst enables oxygen reduction by reacting with the protons generated by the anode, producing water. The ionomer mixed into the catalyst layers allows the protons to travel through these layers.

Gas Diffusion Layers

The GDLs sit outside the catalyst layers and facilitate transport of reactants into the catalyst layer, as well as removal of product water. Each GDL is typically composed of a sheet of carbon paper in which the carbon fibers are partially coated with polytetrafluoroethylene (PTFE). Gases diffuse rapidly through the pores in the GDL. These pores are kept open by the hydrophobic PTFE, which prevents excessive water buildup. In many cases, the inner surface of the GDL is coated with a thin layer of high-surface-area carbon mixed with PTFE, called the microporous layer. The microporous layer can help adjust the balance between water retention (needed to maintain membrane conductivity) and water release (needed to keep the pores open so hydrogen and oxygen can diffuse into the electrodes).

Fuel Cell Systems

To function properly and more effectively, fuel cells need supporting systems that surround them. The design of fuel cell systems is complex and can vary significantly depending upon fuel cell type and application. However, several basic components are found in many fuel cell systems:

- Fuel Cell Stack
- Fuel Processor
- Power Conditioner
- Air Compressor
- Humidifier

Fuel Cell Stack

The fuel cell stack is the heart of a fuel cell power system. It generates electricity in the form of direct current (DC) from electrochemical reactions that take place in the fuel cell. A single fuel cell produces less than 1 V, which is insufficient for most applications. Therefore, individual fuel cells are typically combined in series into a fuel cell stack. A typical fuel cell stack may consist of hundreds of fuel cells. The amount of power produced by a fuel cell depends upon several factors, such as fuel cell type, cell size, the temperature at which it operates, and the pressure of the gasses supplied to the cell.

Fuel Processor

The fuel processor converts fuel into a form usable by the fuel cell. Depending on the fuel and type of fuel cell, the fuel processor can be a simple sorbent bed to remove impurities, or a combination of multiple reactors and sorbents.

Some fuel cells, such as molten carbonate and solid oxide fuel cells, operate at temperatures high enough that the fuel can be reformed in the fuel cell itself. This process is called internal reforming. Fuel cells that use internal reforming still need traps to remove impurities from the unreformed fuel before it reaches the fuel cell. Both internal and external reforming release carbon dioxide, but due to the fuel cells' high efficiency, less carbon dioxide is emitted than by internal-combustion engines, such as those used in gasoline-powered vehicles.

Power Conditioners

Power conditioning includes controlling current (amperes), voltage, frequency, and other characteristics of the electrical current to meet the needs of the application. Fuel cells produce electricity in the form of direct current (DC). In a DC circuit, electrons flow in only one direction. The electricity in your home and workplace is in the form of alternating current (AC), which flows in both directions on alternating cycles. If the fuel cell is used to power equipment that uses AC, the direct current will have to be converted to alternating current.

Both AC and DC power must be conditioned. Current inverters and conditioners adapt the electrical current from the fuel cell to suit the electrical needs of the application, whether it is a simple electrical motor or a complex utility power grid. Conversion and conditioning reduce system efficiency only slightly, around 2%–6%.

Air Compressors

Fuel cell performance improves as the pressure of the reactant gasses increases; therefore many fuel cell systems include an air compressor, which raises the pressure of the inlet air to 2–4 times the ambient atmospheric pressure. For transportation applications, air compressors should have an efficiency of at least 75%. In some cases, an expander is also included to recover power from the high pressure exhaust gasses. Expander efficiency should be at least 80%.

Humidifiers

The polymer electrolyte membrane at the heart of a PEM fuel cell does not work well when dry, so many fuel cell systems include a humidifier for the inlet air. Humidifiers usually consist of a thin membrane, which may be made of the same material as the PEM. By flowing dry inlet air on one side of the humidifier and wet exhaust air on the other side, the water produced by the fuel cell may be recycled to keep the PEM well hydrated.

Hydrogen Fuel Generation / Production

Because hydrogen does not occur naturally in the environment, hydrogen fuel must be derived from other substances that contain hydrogen such asmethanol, gasoline, natural gas, and water. Most hydrogen that is produced today comes from natural gas. If hydrogen is made from water the only byproduct is pure water. If fossil fuels are used as the original source of hydrogen there will be more by-products, such as carbon dioxide. When hydrogen is produced from water electricity is used to split the water molecule. If that electricity comes from a renewable energy source such as wind or solar power, then the resulting hydrogen is a renewable, zero emission fuel.

The processes to extract hydrogen could be divided into four main methods : thermal processes, electrolytic processes, solar driven processes, and biological processes.

Thermal Processes

Thermal processes for hydrogen production typically involve steam reforming, a high-temperature process in which steam reacts with a hydrocarbon fuel to produce hydrogen. Many hydrocarbon fuels can be reformed to produce hydrogen, including natural gas, diesel, renewable liquid fuels, gasified coal, or gasified biomass. Today, about 95% of all hydrogen is produced from steam reforming of natural gas.

Figure 2. Natural Gas Reforming Plant

Electrolytic Processes

Water can be separated into oxygen and hydrogen through a process called electrolysis. Electrolytic processes take place in an electrolyzer, which functions much like a fuel cell in reverse—instead of using the energy of a hydrogen molecule, like a fuel cell does, an electrolyzer creates hydrogen from water molecules.

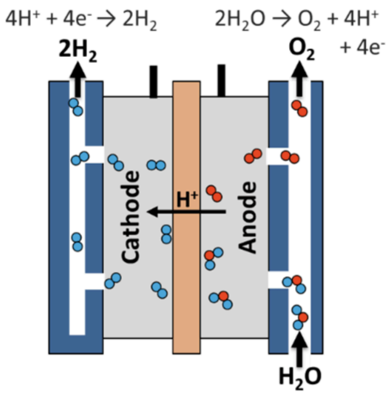

Figure 3. Electrolytic Hydrogen Production Illustration

Solar-Driven Processes

Solar-driven processes use light as the agent for hydrogen production. There are a few solar-driven processes, including photobiological, photoelectrochemical, and solar thermochemical. Photobiological processes use the natural photosynthetic activity of bacteria and green algae to produce hydrogen. Photoelectrochemical processes use specialized semiconductors to separate water into hydrogen and oxygen. Solar thermochemical hydrogen production uses concentrated solar power to drive water splitting reactions often along with

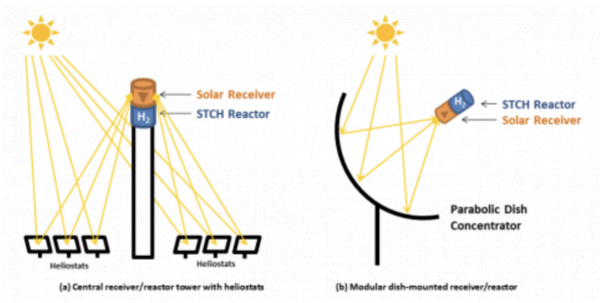

Figure 4. A Thermochemical Water Splitting Method, Driven By Concentrated Sunlight

Biological Processes

Biological processes use microbes such as bacteria and microalgae and can produce hydrogen through biological reactions. In microbial biomass conversion, the microbes break down organic matter like biomass or wastewater to produce hydrogen, while in photobiological processes the microbes use sunlight as the energy source.

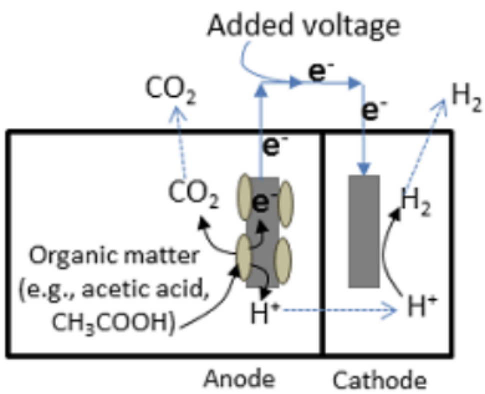

Figure 5. Microbial Electrolysis Cells Illustration, A Microbial Biomass Conversion Device

Hydrogen Fuel Cell Energy Production Efficiency

When considering a fuel cell for an application, considering the efficiency is needed. Fuel cell efficiency directly impacts the operating cost as well as fuel storage requirements. In mobile applications, efficiency becomes even more important because the fuel must be moved with the cell. A less efficient fuel cell must move fuel to travel the same distance, thus wasting some of the generated energy moving the additional fuel.

The table below is retrieved from a book by Ziyad Salameh in 2014 titled Renewable Energy System Design.

Fuel Cell Type	Efficiency
Alkaline	70%
Molten carbonate	60–80% with cogeneration
Phosphoric acid	40–80% with cogeneration
Solid oxide	60%
Proton exchange membrane	40–50%

Figure 6. Tab1e of Fuel Cells Efficiency Based On The Electrolyte Types

It is important to notice that the efficiency range for the fuel cells range from 40 to 80%. One major advantage of fuel cells becomes apparent when we compare their efficiencies to the efficiency of the internal combustion engine. The typical efficiency of an internal combustion engine is on the order of 20%. Thus, to power a vehicle with the same performance and range, we only need to carry half as much fuel, assuming a 40% efficient PEM cell stack.

Measuring the efficiency of a fuel cell is simply the practical cell-output voltage divided by the thermodynamically reversible voltage at the stated temperature and pressure of operation. The formulas may vary between the type of fuel cells and the type of research that covers it, but most follow this same underlying concept.

A book by Zhang, Jiujun et. al; 2013 titled PEM Fuel Cell Testing and Diagnosis, present the formula below to realistically measure the efficiency of a fuel cell, by factoring in other variables and processes besides the energy output and input itself.

$$\eta_e = \frac{nFV_{\mathrm{cell}}}{\Delta H} \frac{1}{\lambda} \frac{I_{\mathrm{cell}}}{I_{\mathrm{cell}} + i_{\mathrm{loss}}} \times 100\%$$

Figure 7. Hydrogen Fuel Cells Efficiency Formula

For example, at room temperature, if a H_2/air fuel cell is operated at a current of 100 A (i = 100 A) and a cell voltage of 0.6 V, and the hydrogen stoichiometry is 1.2, and the hydrogen loss current is 5 A (i_{loss} = 5A), then the calculated fuel cell efficiency according to Eqn (1.75) is only about 35.5% (with ΔH = 285.8 kJ mol^{-1} and n = 2), which is much smaller than the theoretical electrical energy efficiency of 83%.

Hydrogen Fuel Cells Applications

A typical hydrogen fuel cell produces 0.5 V to 0.8 V per cell, which is insufficient for most applications. Therefore, To increase the voltage individual cells can be connected in series. This arrangement is called a fuel cell stack. The cross sectional area of a fuel cell affects its ability to produce current. Greater area means more reaction sites, and this allows more current to be generated. Current voltage equals power. So, by stacking cells in series to build voltage and increasing cell area to boost current, it is possible to generate very large amounts of electrical power.

With the portability of fuel cells and its ability to be stacked, hydrogen fuel cells could be applied everywhere. There have been demonstrations of applying fuel cells onto handheld devices, vehicles, stationary power for buildings, power for remote areas, in the military, space crafts, submarines, and many more. These are the potential of what hydrogen fuel cells could be applied with.

A review paper of fuel cells development by Fan, Lizin; Tu, Zhengkai; Chan, Siew in 2021 shows that Component manufacturing is the highlight in fuel cell commercial application, and Ballard, LCS, Toyota, and SRM are the major component suppliers in the world. There are some companies that provide the mature design, development, production, and service of PEMFCs, such as Ballard, Plug Power, and Hydrogenics Corporation, and the product of the latter, called HyPM, can output 1 KW–1 GW electricity. Bloom Energy and UTC Power once worked for NASA for energy supply of spacecraft, and are now focusing on the development of SOFCs to provide electricity in microhousehold fuel cells by using their marquee products, Bloom Box and Pure Cell, respectively. Fuel Cell Energy is mainly engaged in the research of stationary fuel cells and its main products are MCFCs for on-site power generation, cogeneration, and distributed generation used in the power industry, business and enterprise, government agencies, and so on. The company's current DFC power plant has produced ultraclean, efficient, and reliable power in more than 50 locations worldwide, with the ability to produce more than 1.5 billion kWh of ultraclean power and more than 300 MW of generating capacity. In October 2009, Toshiba released a direct methanol fuel cell (DMFC) called Dynario in Japan, which automatically generates electricity as long as methanol is fed into the unit. The power generated can be used to charge a built-in lithium-ion battery or provide a DC 5 V output via a USB interface; however, the Dynario provides a slightly smaller current of 400 mA. The Dynario machine itself has 14 ml of methanol content, which can provide sufficient power to charge two mobile phones. The UK's ITM Energy Company implemented a hydrogen energy demonstration project in the Rotherham area of the UK to provide partial power support and hydrogen fuel for nearby buildings. The system consists of a 225 kW wind turbine, an advanced electrolysis cell, a hydrogen storage system for storing 200 kg of hydrogen, and a fuel cell power system with a power of 30 kW. The hydrogen energy integrated demonstration system combines ITM Energy's most advanced hydrogen energy commercialization module and is a very important component of the UK's hydrogen energy infrastructure. Sweden's Vatten Fall has established Europe's first hydrogen–wind energy renewable energy power plant, which uses wind energy to produce hydrogen. Taking China as an example to introduce component production, stack companies are supported by independent research and development and technology introduction, such as Guangdong Guorui, Weichai. MEA, PEM, and BP can be mass-produced, whereas the catalyst and GDL can only be produced at small scales. University related high-tech enterprises are the main force for these productions, such as Dongyue or Wuhan University of Science and Technology

III. Conclusion

Three main conclusions are drawn after reviewing related papers, articles, covering the fuel cell design, and commercial markets.

(1) In the broader terms, Hydrogen fuel cells are a viable solution and alternative to fossil fuels, with their true zero emission and highly efficient power source. There are still major drawbacks preventing it from being used at a global scale. But, as long as investments and developments are continued, in the future hydrogen fuel cells could deliver their full potential while being cost effective.

(2) Stability and cost are still the main challenges in commercial releasing, which requires the technical support for large-scale, low-cost, and stable applications. Material, structural design, and system integration are the three key factors limiting the energy improvement and technological advancement. The structure of fuel cells should be designed to perform more efficiently, and hydrogen technologies in generation and storage must focus on achieving the material advancements for the development of its safety and stability during operation. Systems should be properly designed to link between hydrogen production, storage, transportation, and utilization with durability and high efficiency.

(3) Successful applications need more powerful, focused, and long-term energy strategies and public policies with great pressure on challenging the traditional energy system, requiring low-carbon technology innovation. Regional coordination organization should make proper policies considering the imbalance between different countries, and offer support and guide countries in hydrogen policy-making. For a country, a fully developed policy system should be founded in advance to support the innovation and development of hydrogen and fuel cell technology that provides the stability and vitality to satisfy feasible and reasonable performance and cost targets.

Appendix

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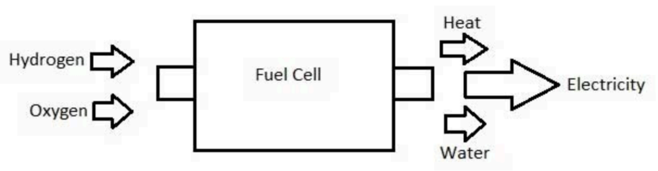
FIGURES



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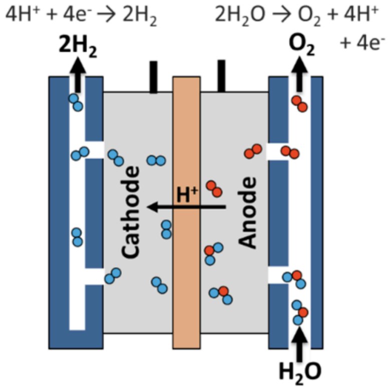

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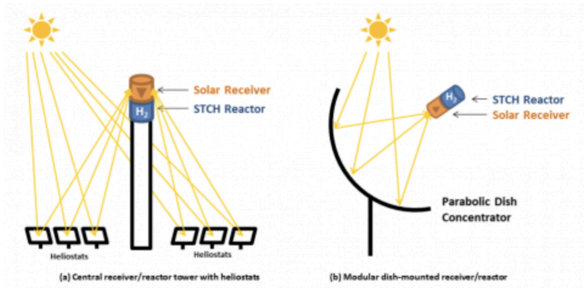

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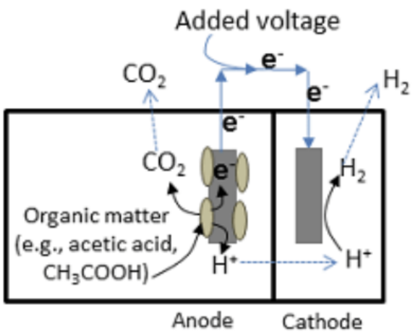

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